

Cross-Tool Practice Set — Questions

Dataset: `sales_practice.csv` — 6,040 rows, 19 columns, retail order transactions. Also supplied as `sales_practice_excel.xlsx` with a `Data` sheet (raw) and a `DataClean` sheet (duplicates removed).

Columns: OrderID, OrderDate, Region, City, SalesRep, Manager, Category, SubCategory, ProductName, Channel, CustomerSegment, CustomerID, Units, UnitPrice, DiscountPct, UnitCost, ShipDays, Returned, PaymentMethod

Derived measures used throughout — define these first in every tool:

```
NetRevenue = Units * UnitPrice * (1 - DiscountPct)
TotalCost   = Units * UnitCost
GrossMargin = NetRevenue - TotalCost
```

Section A is a cleaning stage and runs on the RAW data. Everything from Section B onward runs on the **deduplicated** data. Do Section A first — the answers depend on it. In Excel that means working on `DataClean` (or running Data > Remove Duplicates yourself); in SQL and Spark, deduplicate as your first step.

How to use this. Answer every question in **all three tools** — Excel, SQL and PySpark — and confirm the numbers match. When they don't, one of them is wrong, and finding out which is the actual exercise. Sections F and G are Excel-only.

Section A — Data Quality (run on RAW data)

A1. How many rows are in the raw file? **A2.** How many rows remain after removing exact duplicate rows? **A3.** How many rows have negative `Units`? **A4.** How many rows have a blank `ShipDays`? **A5.** How many rows have a blank `CustomerSegment`? **A6.** List every distinct text variant appearing in `Region`. How many are there, and what are the two separate defects? **A7.** How many rows are involved in a duplicated `OrderID`? **A8.** Write the expression that produces a cleaned `Region` column with exactly four values. **A9.** Decide and justify: how would you handle the negative `Units` rows, the blank `ShipDays`, and the blank `CustomerSegment`? Would you use the same strategy for all three?

Section B — Basic Aggregation

B1. Total net revenue across the whole dataset. **B2.** Total net revenue by region, sorted descending. **B3.** Total net revenue and order count by category, sorted by revenue descending. **B4.** Total units sold. **B5.** Net revenue for the South region only. (Careful — the `Region` column is dirty.) **B6.** Net revenue for Online channel + Corporate segment + year 2025 only. **B7.** Average units per order for Electronics, excluding returned orders. **B8.** How many distinct sales reps operate in the South region? **B9.** Largest single-order net revenue in each region. **B10.** How many orders carried any discount at all? **B11.** How many distinct customers appear in the data?

Section C — Business Metrics

C1. Return rate (proportion of orders returned) by category. **C2.** Gross margin percentage by category, sorted descending. Which category is most profitable? **C3.** Average `ShipDays` by channel. Does your answer treat blanks correctly? **C4.** Top 5 sales reps by net revenue. **C5.** Revenue by region and year, as a cross-tab. **C6.** Which three calendar months had the highest revenue? What does that suggest? **C7.** Compare discounted vs non-discounted orders: count, average units, total revenue. Did discounting increase units per order? **C8.** Bucket `ShipDays` into 1-3, 4-6, 7-9, 10+ and `Unknown`, and count orders in each. **C9.** Revenue difference between 2025 and 2024.

Section D — Window Functions & Ranking

(SQL and PySpark; in Excel use a pivot with a value filter or helper formulas.)

D1. Top 3 products by revenue **within each category**. **D2.** Rank sales reps by revenue **within their region**. Who is number one in each? **D3.** Month-over-month revenue growth percentage across the whole dataset. **D4.** Running cumulative revenue by month for the North region. **D5.** Each category's revenue as a percentage of total revenue. **D6.** How many products earn more revenue than the average product in their own category? **D7.** How many customers placed more than 10 orders? **D8.** Are there any sales reps with zero returned orders?

Section E — Explain & Compare

(No code — these are reverse-KT style. Answer aloud.)

E1. You computed B5 with a simple `SUMIF` / `SUMIFS` on "South" and got a smaller number than expected. Explain exactly why. **E2.** Why must the deduplication happen before Section B and not after? **E3.** In Excel, `COUNTIF` reports 8 distinct `Region` variants but SQL reports 12. Explain the discrepancy. **E4.** C3 asks about blanks. Explain what `AVERAGEIFS` does with blank cells, and what `AVG()` does with NULLs in SQL. Are they the same? **E5.** For D1, explain why `ROW_NUMBER` is the right choice rather than `RANK`. **E6.** You need this report refreshed monthly on 50 million rows. Which of the three tools do you use, and why?

Section F — Pivot Tables (Excel only)

F1. Build a pivot: Region in rows, Category in columns, sum of NetRevenue as values. Format as currency with no decimals. **F2.** Modify it to show each cell as a percentage of its **row** total. **F3.** Build a pivot of revenue by Year and Quarter using **date grouping**, not a helper column. **F4.** Build a pivot with Category in rows and both Units and NetRevenue as values, side by side. **F5.** Add a **Calculated Field** for Gross Margin %. Explain what it is actually computing. **F6.** Show the top 5 sales reps by revenue using a **value filter**, not by sorting and deleting rows. **F7.** Group `ShipDays` into buckets using pivot **numeric grouping**. **F8.** Add slicers for Region and Channel, and a timeline on OrderDate, connected to two different pivots at once. **F9.** Use `GETPIVOTDATA` to pull South region revenue into a cell outside the pivot. **F10.** Double-click a value cell to drill through. What does Excel generate, and when is this useful? **F11.** Change a `UnitPrice` in the source, refresh, and confirm the pivot updates. Then add a new row and refresh again. What happens, and why? **F12.** Explain the difference between a Calculated Field and a Calculated Item, with an example of each from this dataset.

Section G — Charts & Dashboard (Excel only)

G1. Line chart of monthly revenue over the full period. **G2.** Column chart of revenue by category, sorted descending. **G3.** Combo chart: revenue as columns and return rate as a line on a secondary axis, by category. **G4.** Scatter plot of DiscountPct against Units. Is there a visible relationship? **G5.** Conditionally format the top 10% of orders by net revenue. **G6.** Add a sparkline column showing each region's monthly revenue trend. **G7.** Build a one-page dashboard: three KPI cards (total revenue, total orders, return rate), three charts, and slicers controlling all of them. Hide the working sheets. **G8.** Which chart type would you use to show each region's share of total revenue, and why not a pie chart if there were 12 categories?